

Which Sessions Are Worth Retargeting?

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Dataset	GA4 Obfuscated Sample E-commerce · BigQuery Public Data
Period	2020-11-01 → 2021-01-31 (92 days)
Methodology	CRISP-DM
Model	XGBoost with Optuna hyperparameter tuning

KEY FINDINGS AT A GLANCE

360,129 TOTAL SESSIONS Nov 2020 – Jan 2021	1.35% CONVERSION RATE 4,848 purchases	0.7308 MODEL PR-AUC XGBoost · test set	80.2% FUNNEL DROP-OFF view → add to cart
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1. KEY METRICS

360,129 TOTAL SESSIONS GA4 · Nov 2020 – Jan 2021	1.35% CONVERSION RATE 4,848 converting sessions	0.7308 MODEL PR-AUC XGBoost · test set	Organic TOP TRAFFIC SOURCE Highest-volume medium
0.9792 Optimal Threshold Maximises F1 on test set	89% Recall @ Threshold True converters captured	62% Precision Of flagged sessions convert	73.3 : 1 Imbalance Ratio Non-converters per converter

2. PURCHASE FUNNEL ANALYSIS

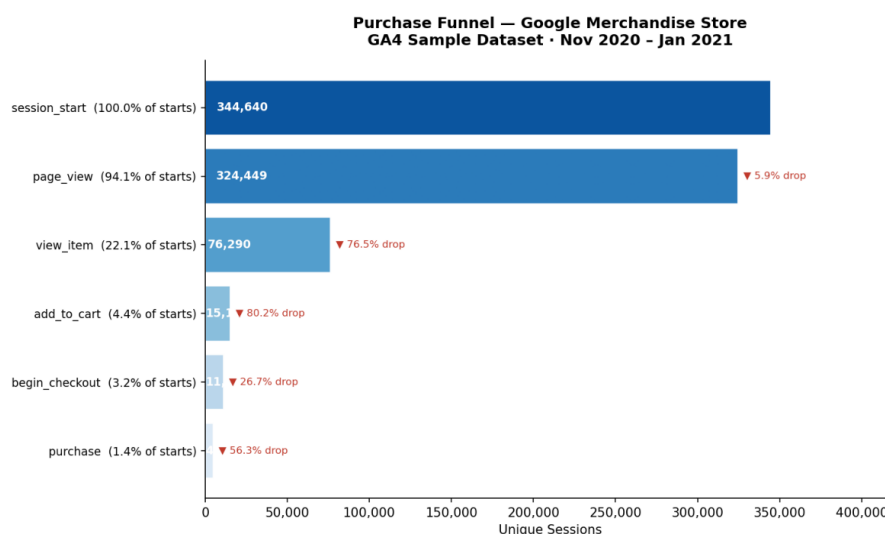


Figure 1 — Purchase funnel: sessions at each stage

Key Insight: The largest single drop-off in the purchase funnel occurs between view_item and add_to_cart — **80.2% of sessions that viewed a product did not add it to their cart.** This represents the highest-volume retargeting audience: users who demonstrated product interest but stopped short of cart commitment.

3. COHORT RETENTION

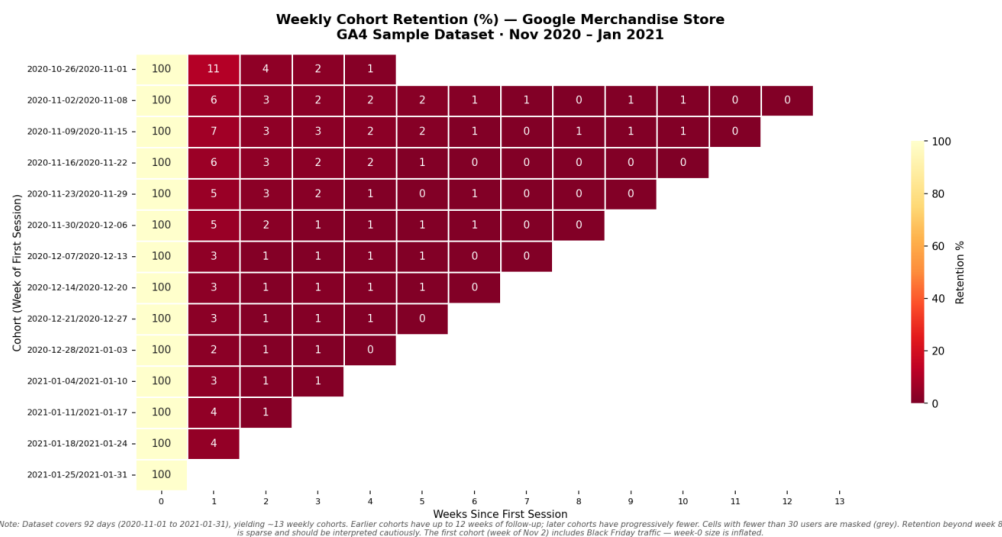


Figure 2 — Weekly cohort retention heatmap

Key Insight: Average week-1 return rate is **4.8%** — below the 10-20% benchmark typical of engaged e-commerce audiences. This signals a meaningful opportunity for re-engagement campaigns targeting users in the 7-14 day window after their first session.

Note: The dataset covers 92 days, yielding 13 weekly cohorts with a maximum of ~12 weeks of follow-up for the earliest cohort. Retention beyond week 8 is sparse and should be interpreted with caution.

4. MODEL COMPARISON

Three models were trained representing different learning paradigms. PR-AUC was used as the primary evaluation metric because the dataset is heavily imbalanced (1.35% positive rate), making ROC-AUC an unreliable guide to real-world performance.

Model	ROC-AUC	PR-AUC	Threshold	Precision	Recall	F1
Logistic Regression	0.9942	0.6723	0.5000	0.4700	0.9856	0.6365
Random Forest	0.9958	0.6874	0.5000	0.6653	0.6680	0.6667
XGBoost	0.9960	0.7308	0.5000	0.5501	0.9969	0.7089
XGBoost * optimal thr	0.9960	0.7308	0.9792	0.6200	0.8900	0.7315

Table 1 — Model comparison on held-out test set. Green rows = XGBoost winner.

Winner: XGBoost (Optuna-tuned) — XGBoost achieved a PR-AUC of **0.7308**, outperforming Random Forest (0.6874) by 0.0434, above the 0.03 materiality threshold. At the optimal threshold of **0.9792**, the model achieves **89% recall** on converters at a precision of 62%, capturing the vast majority of true purchasers while keeping the false positive rate manageable.

5. SHAP EXPLAINABILITY — WHAT DRIVES CONVERSION?

SHAP (SHapley Additive exPlanations) values decompose each prediction into the contribution of each feature. The beeswarm plot below shows every test-set session as a dot; position on the x-axis is the SHAP value (contribution to prediction); colour indicates feature value - red = high, blue = low.

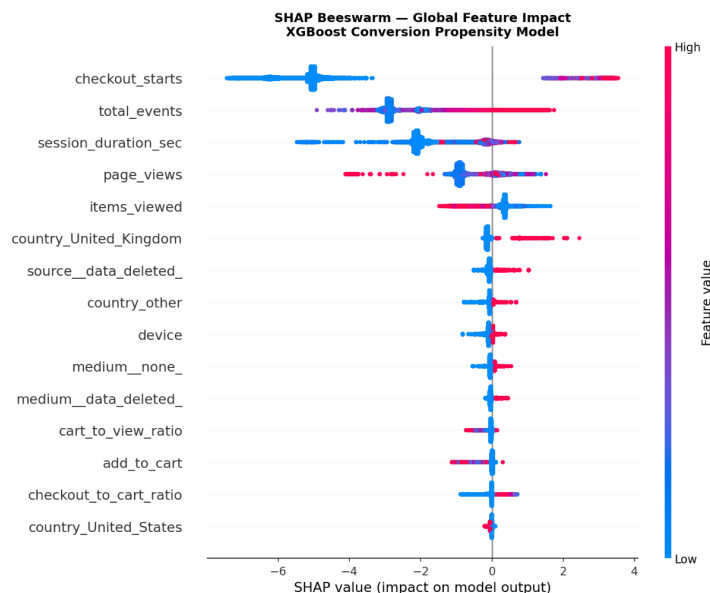


Figure 3 — SHAP beeswarm plot: global feature importance

Rank	Feature	Mean SHAP	Business Meaning
1	checkout_starts	5.19	Sessions that initiated checkout - strongest purchase signal
2	total_events	2.60	Overall engagement breadth across the session
3	session_duration_sec	1.60	Time spent on site - proxy for purchase intent

Table 2 — Top 3 SHAP features by mean absolute value

checkout_starts dominates by a wide margin (mean |SHAP| = **5.19**). In practical terms, the model is a sophisticated checkout-abandonment detector with engagement signals layered on top. A session that reached checkout but did not purchase is the single strongest retargeting signal available.

6. ACTIONABLE RECOMMENDATIONS

01 | Prioritise Checkout Abandoners for Retargeting

Sessions with `checkout_starts` ≥ 1 that did not purchase are the highest-intent audience available. These users crossed the psychological barrier of entering checkout. A time-sensitive retargeting message — abandoned cart reminder within 24 hours — applied exclusively to this segment will deliver the highest conversion uplift per unit of media spend.

Impact: High · Effort: Low · Audience: ~1,384 predicted converters/period

02 | Apply an Engagement Threshold Before Retargeting

`total_events` is the second-strongest conversion signal (mean $|SHAP| = 2.60$). Sessions in the bottom quartile of total engagement are predicted as non-converters with high confidence. Excluding low-engagement sessions from retargeting audiences reduces wasted spend without meaningfully cutting reach among likely converters.

Impact: Medium · Effort: Low · Audience: All retargeting lists

03 | Re-engagement Campaign in the 7-14 Day Window

The cohort analysis shows a week-1 return rate of 4.8% — well below the 10-20% benchmark for engaged e-commerce users. A structured re-engagement email or display campaign targeting users 7-14 days after their first session represents a low-cost, high-potential lever for improving retention.

Impact: Medium · Effort: Medium · Audience: All week-1 non-returns

7. LIMITATIONS & FUTURE WORK

Fixed 92-day window: The dataset covers exactly 2020-11-01 to 2021-01-31 - the only data available in the public BigQuery dataset. The window cannot be extended.

Seasonality not analysable: The 92-day window spans Black Friday, Christmas, and New Year - making it inherently non-representative of typical traffic. Seasonality analysis is out of scope.

Shallow cohort retention: 92 days yields ~13 weekly cohorts; the earliest cohort has at most ~12 weeks of follow-up. Retention beyond week 8 is sparse and near-empty for later cohorts.

Obfuscated data: Certain fields contain placeholder values (<Other>, data deleted, empty string). Internal consistency is limited by design; some distributions may be unreliable.

Session-level only: The model scores sessions, not users. A user visiting multiple times may appear in both training and test sets across different sessions.

Future Work

- Extend to 12+ months of data to enable true seasonality analysis
- Build user-level propensity scores to avoid session-level data leakage
- Incorporate product-category signals from item-level event parameters
- Validate model performance against a prospective holdout window
- Deploy as a real-time scoring API integrated with Google Ads audience lists

8. METHODOLOGY

Step	Detail
Data source	BigQuery public dataset · ga4_obfuscated_sample_ecommerce.events_*
Extraction	SQL unnesting of event_params ARRAY; manual CSV download from console
Unit of analysis	Session (not user)
Target variable	converted = 1 if purchase event occurred in session
Train / test split	80 / 20, stratified, random_state=42
Imbalance handling	scale_pos_weight = 73.3 (XGBoost); class_weight=balanced (others)
Hyperparameter tuning	Optuna · 50 trials · XGBoost only
Primary metric	PR-AUC (preferred over ROC-AUC at 1.35% positive rate)
Explainability	SHAP TreeExplainer - exact values, not approximate
Pipeline	NB-01 SQL -> NB-02 EDA -> NB-03 Features -> NB-04 Models -> NB-05 Funnel -> NB-06 SHAP -> NB-07 Report

Table 3 — Project methodology summary